Project Proposal

Names:

Luna Kim

James Giovinazzo

Lyba Naseem

Student Numbers:

400521674

400583359

400602518

Course Code:

PROTECH 2SE3 Systems Engineering Fundamentals

Course Section:

C01, Group 9

Date of Submissions:

January 25th, 2026

Automated Coin Sorting and Counting System Description

General System Overview

Automated coin sorting and counting system is designed to recognize, sort, and count mixed coins without requiring human involvement. It is used in commercial and financial environments where efficiency and accuracy are important. The system reduces manual handling and operational errors while maintaining safe and reliable operation. It continuously monitors operating conditions and responds in a controlled manner. It is also designed to operate safely during power outages and to make routine maintenance easier.

Scenario 1: Operation

The system accepts mixed coins from the user, detects their presence, and initiates the automated coin sorting and counting process. Once the operation is complete, the system displays the final results to the user and enters an idle state while awaiting the next operation.

Scenario 2: Processing Behaviors

The system shall continuously process coins during operation, classify coins, and route them to appropriate internal storage areas. Throughout processing, the system updates the total coin count and monetary value while monitoring operating conditions to ensure normal operation.

Scenario 3: Validation and Routing

The system will validate the coinage by determining if it is a genuine currency and not a foreign object. The system would then direct the coins into its allotted storage. The system will then increase the count and total monetary value in real-time as each coin reaches its destination.

Scenario 4: Batch Completion

The system will detect when no further coins enter processing during any given time. A summary will be created displaying the count per allotted coin and the total value of the sum of coins. The system would then have a digital record and reset the session counter for the next user.

Scenario 5: Exception (Jamming)

The system will monitor the flow of coins through internal instruments. If a blockage is detected, the system will halt intake to prevent any damage or further cloggage. Then, the system will attempt to clear the jam before notifying an operator of a manual jam. Damaged coins will be routed to a separate storage area for user retrieval.

Scenario 6: Capacity Management for Overflow

The system monitors the fill level of each internal storage container. The system will alert an operator when any allotted storage is in full capacity. If a container reaches max capacity, the system will prevent more intake of that specific denomination by routing it to a separate storage area for user retrieval, until the bin is empty.

Scenario 7: Maintenance Monitoring

The system will track the total number of coins processed to schedule routine cleaning in internal paths (kind of like an oil change). The system will monitor health of integral movement components and flags automatically diagnosed degradation in speed or precision. Automatic notifications will appear to the operator if the system requires maintenance if it cannot automatically fix itself.